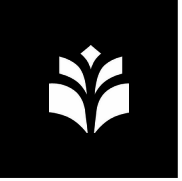
Accessibility, components and Bootstrap

Web programming I

Course materials originally by Sonsoles López-Pernas, adapted by Maria Kuismin

Accessibility, components and Bootstrap

Student is able to
design and implement interactive and responsive web pages using HTML, CSS, JavaScript, and Bootstrap.
analyze user interface requirements and design reusable, component-based web applications using React



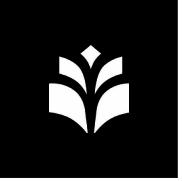
Content

Accessibility
- What is web accessibility, and how does it benefit everyone?
- What are the main principles of accessible design?

UI components and frameworks
- What are UI elements, and why are they used consistently?
- What is a UI component library, and why is it useful?

Bootstrap
- What is Bootstrap, and how do we add it to a project?
- How do we use Bootstrap's grid and components?
- How can Bootstrap be customized?

Alternative frameworks
- What other CSS frameworks exist beyond Bootstrap?



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Web accessibility

- Accessibility means making websites usable for all people, including those with disabilities.
- Disabilities can be visual, auditory, motor, or cognitive.
- Accessible design benefits everyone, e.g., better contrast helps in bright sunlight, captions in noisy environments, keyboard navigation those who prefer keyboards over mice

Why accessibility matters

- **Legal requirements** e.g., WCAG (Web Content Accessibility Guidelines), EU accessibility act
- **Social responsibility**: equal access to information and services.
- **Practical benefits**: improves usability, SEO (Search engine optimization), and overall user experience.

Key principles of accessible design

- **Perceivable** – information must be visible or audible.
- **Operable** – users can navigate with mouse, keyboard, or assistive tech.
- **Understandable** – content and navigation are clear.
- **Robust** – works well with different devices and assistive technologies.

Examples of accessibility features

- Text alternatives for images (alt attribute).
- High color contrast between text and background.
- Captions or transcripts for videos.
- Keyboard navigation (tabbing through links and forms).

ARIA attributes improve accessibility

- ARIA = *Accessible Rich Internet Applications.*
- A set of HTML attributes that improve accessibility for people using assistive technologies (like screen readers).
- They don't change how a page looks, but they add extra meaning for assistive software.

```
<button aria-expanded="false" aria-controls="menu1">
  Toggle Menu
</button>
```

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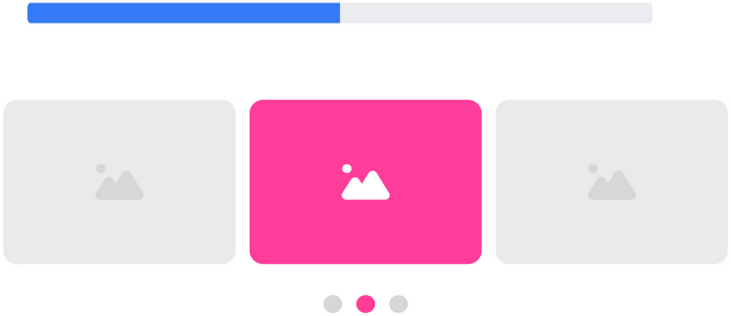
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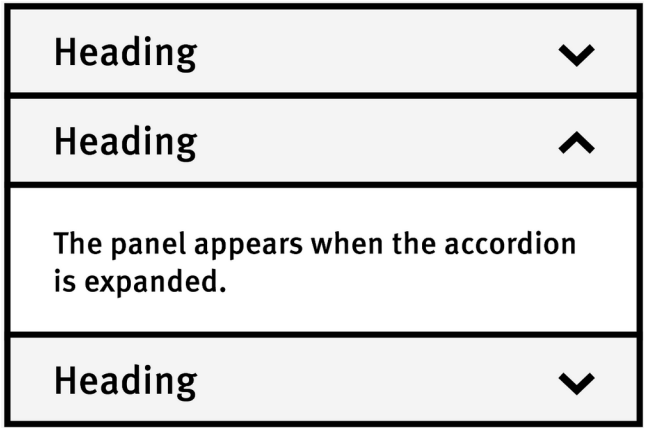
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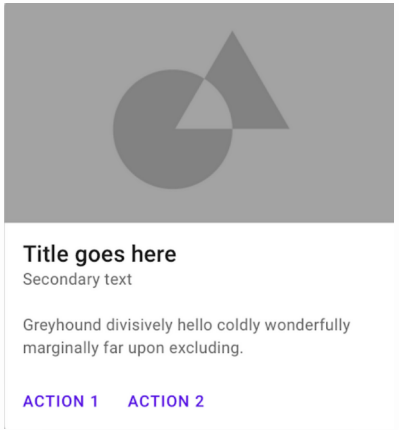
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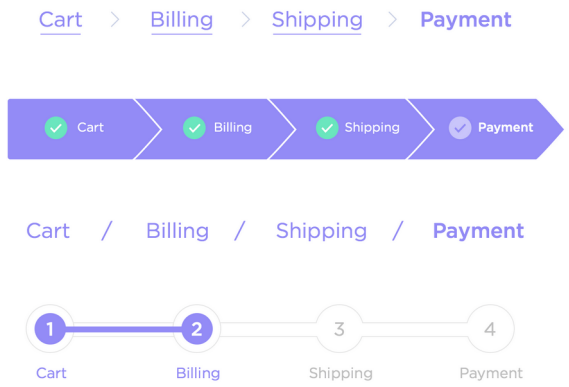
Web User Interface components

- UI elements are the building blocks of apps and web sites.
- They are what users interact with when they are using the product.

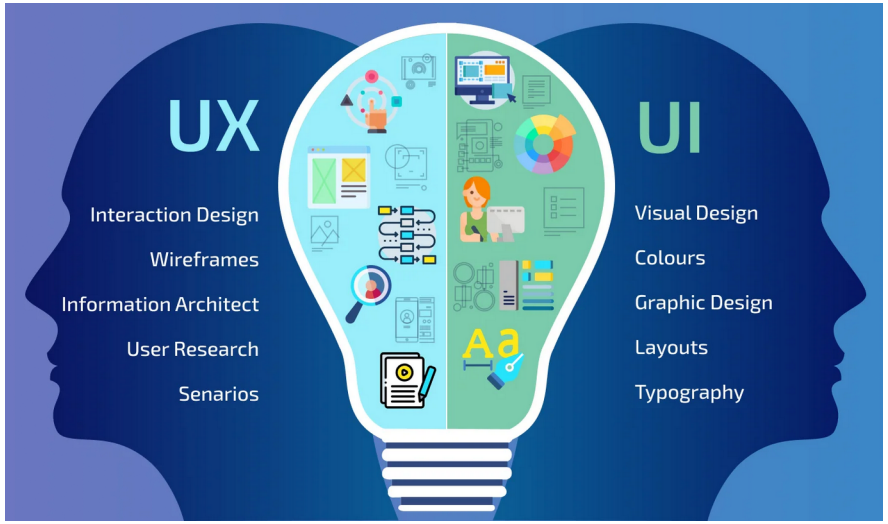








UI & UX

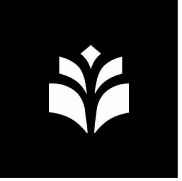


- UX, User eXperience
  = how users experience a product
- Familiar design patterns build user trust, breaking patterns confuses users
- Designers reuse well-known UI components to keep experiences predictable
- This is why the same components appear across many websites

Web UI component libraries

A collection of reusable components, guided by clear standards

For consistency, scalability, and efficiency in web development



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Bootstrap framework

- An open-source CSS framework
- Provides ready-made HTML, CSS, and JavaScript components
- Designed for responsive, mobile-first websites
- Free and very widely used

https://getbootstrap.com/



Benefits of using framework like Bootstrap

- Speeds up development
- Responsive design made easy
- Built-in components (navigation bars, modals, etc.)
- Consistency across devices and browsers
- Large community and documentation

Adding Bootstrap to a Project

a) Using CDN

```
<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min
.css" rel="stylesheet" integrity="sha384-
sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB"
crossorigin="anonymous">
```

b) Downloading and including locally

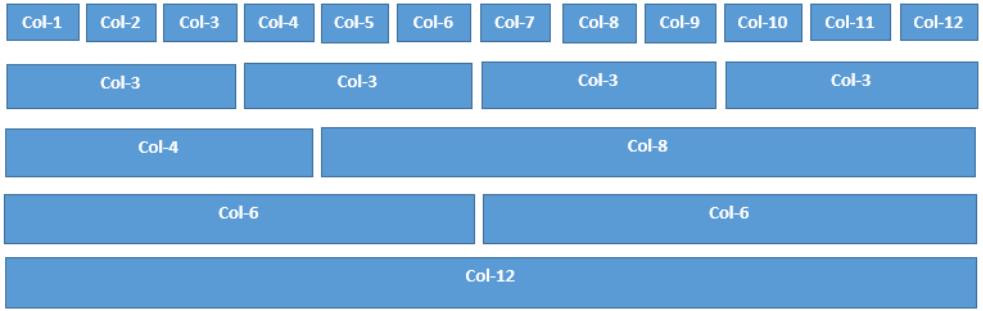
How Bootstrap works: utility classes

Bootstrap works by adding CSS classes to HTML elements:

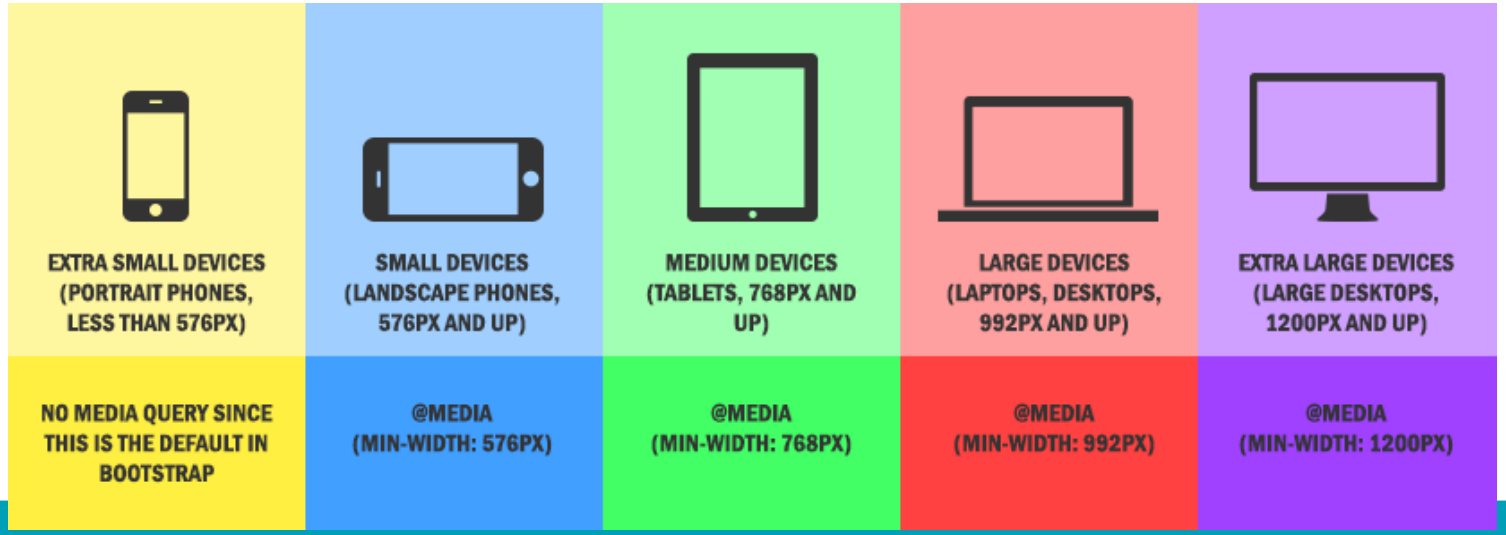
```
<button class="btn btn-primary">Submit your application</button>
<p class="text-success">Application sent successfully</p>
<div class="bg-warning">Please fill in all required fields</div>
```

- "btn", "text-*", "bg-*" are pre-defined Bootstrap classes
- Each class applies a set of CSS rules
- You combine classes to achieve the desired look
- No need to write custom CSS for common patterns

Grid System



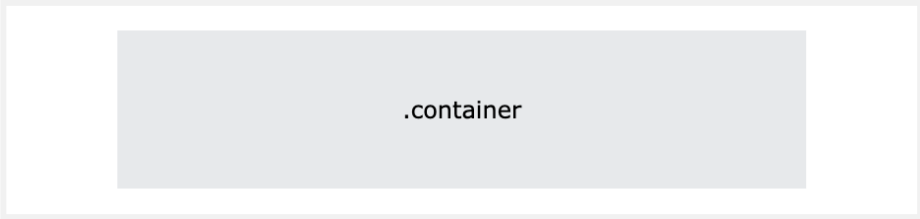
- Mobile-first, 12-column layout
- Classes for different breakpoints: **.col-xs-** for extra small devices (≤ 576px), **.col-sm-** for small devices (≥576px), **.col-md-** for medium devices (≥768), **.col-lg-** for large devices (≥992px), **.col-xl-** for extra large devices (≥1200px)

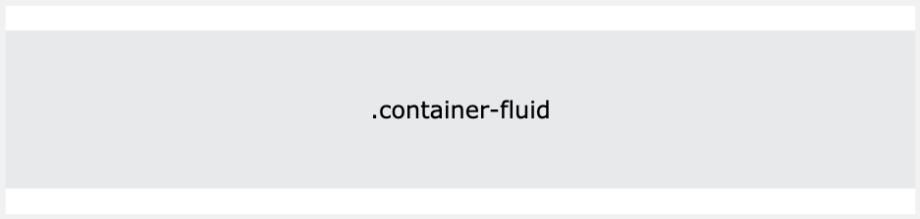


Containers

Containers are used to pad the content inside of them, and there are two container classes available:

1. The `.container` class provides a responsive fixed width container
2. The `.container-fluid` class provides a full width container, spanning the entire width of the viewport

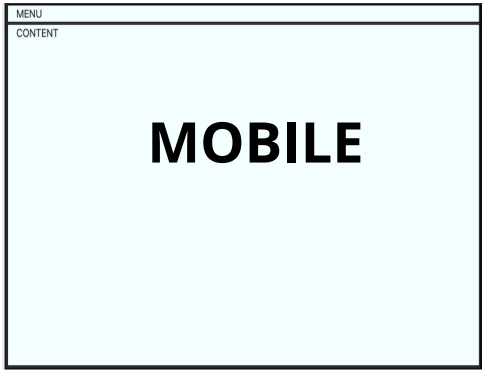


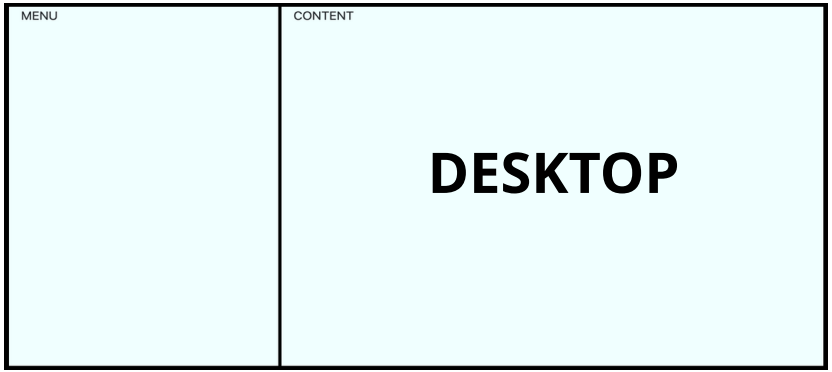


Layout - desktop and mobile

```
<div class="container-fluid">
    <div class="row">
        <div class="col-12 col-lg-4">MENU</div>
        <div class="col-12 col-lg-8">CONTENT</div>
    </div>

</div>
```





https://appbrewery.github.io/bootstrap-layout

Bootstrap Components

Pre-built UI elements

- Navigation bars
- Buttons
- Forms
- Modals
- Carousels
- Icons
- etc.



carousel



icons

Component: Navbar

```
<nav class="navbar navbar-expand-sm bg-primary navbar-dark">
  <ul class="navbar-nav">
    <li class="nav-item active">
      <a class="nav-link" href="#">Active</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Link</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Link</a>
    </li>
    <li class="nav-item">
      <a class="nav-link disabled" href="#">Disabled</a>
    </li>
  </ul>
</nav>
```

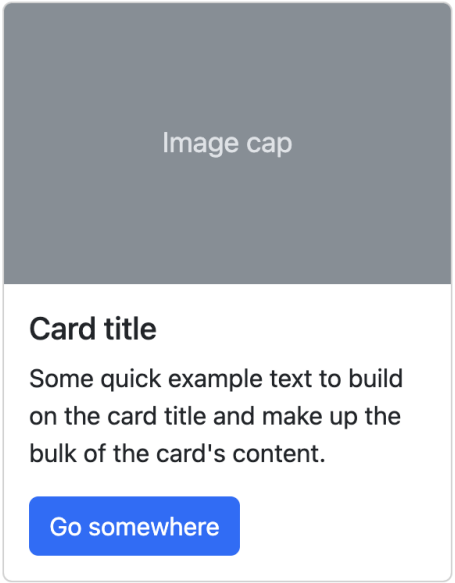


Component: Card

A flexible content container.
Options: headers, footers, contextual colors, display options

Key Bootstrap classes
- card (the container)
- card-img-top (image at top)
- card-body (content area)
- card-title (title)
- card-text (text content)

.



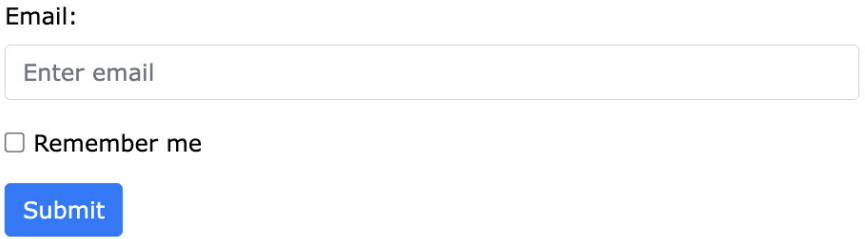

```
<div class="card">
  <img class="card-img-top" alt="..." src="..." >
  <div class="card-body">
    <h5 class="card-title">Card title</h5>
    <p class="card-text">Some quick example text to build on
the card title and make up the bulk of the card's content.</p>
    <a href="#" class="btn btn-primary">Go somewhere</a>
  </div>

</div>
```

Component: Form

Key Bootsrap classes:
- form-group (wrap each field)
- form-control (styles inputs and selects)
- form-check (for checkboxes and radios)
- form-check-input (the checkbox itself)
- form-check-label (the label)

```
<form action="/action_page.php">
  <div class="form-group">
    <label for="email">Email address:</label>
    <input type="email" class="form-control" placeholder="Enter email"
id="email">
  </div>
  <div class="form-group form-check">
    <label class="form-check-label">
      <input class="form-check-input" type="checkbox"> Remember me
    </label>
  </div>
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```



https://www.w3schools.com/bootstrap4/bootstrap_forms_inputs.asp

Component: Breadcrumb

Shows the current page's location in a navigation hierarchy.
CSS automatically adds separators between items.

Key Bootstrap classes:
- breadcrumb (the list)
- breadcrumb-item (each level)
- active (the current page)

Home / Library

Home / Library / Data

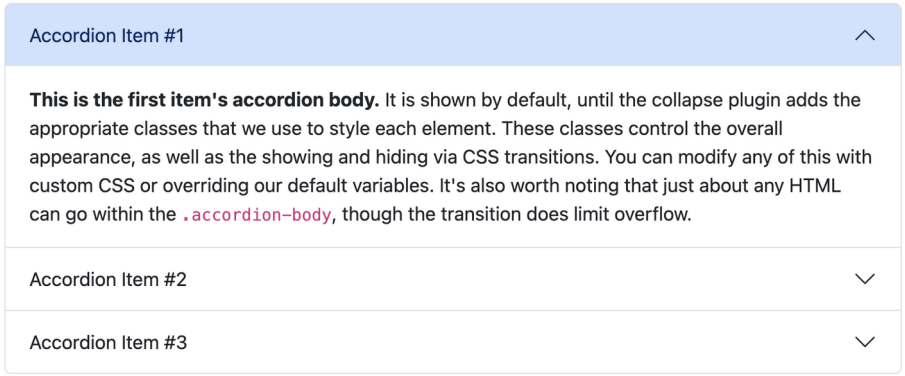
```
<nav aria-label="breadcrumb">
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item active" aria-current="page">Library</li>
  </ol>
</nav>

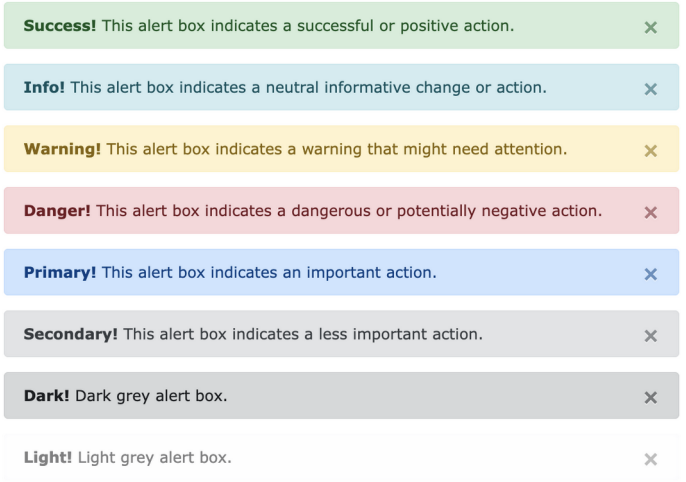
<nav aria-label="breadcrumb">
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item"><a href="#">Library</a></li>
    <li class="breadcrumb-item active" aria-current="page">Data</li>
  </ol>
</nav>
```

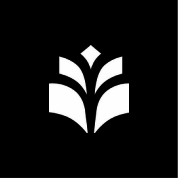
More Bootstrap components

Beyond cards, navbars, and forms, Bootstrap offers accordion (collapsible sections), alert (status messages), badge (small labels), button (various styles), carousel (image slideshow), dropdown (menu), modal (popup dialog) and many more

Full list and documentation: https://getbootstrap.com/docs/5.3/components/







Customizing Bootstrap: Themes

Bootswatch (bootswatch.com) offers free Bootstrap themes

- Different visual styles without changing your HTML
- Simply replace the Bootstrap CSS link with a theme link

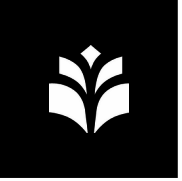
original Bootstrap:

```
<link href="https://cdn.../bootstrap.min.css" rel="stylesheet">
```

e.g. with Bootswatch Lumen theme:

```
<link href="https://bootswatch.com/5/lumen/bootstrap.min.css"
rel="stylesheet">
```

The same HTML classes still work, but the look changes.



Customizing Bootstrap: Your own CSS

- override Bootstrap's default styles with your own CSS.
- Add your custom stylesheet after Bootstrap:

```
<link href="bootstrap.min.css" rel="stylesheet">
<link href="my-custom.css" rel="stylesheet">
```

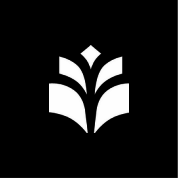
- In my-custom.css, override Bootstrap classes:

```
.btn-primary {
  background-color: #6f42c1; /* Custom purple */
  border-color: #6f42c1;
}
```

- Remember the CSS cascade: later rules override earlier ones

Bootstrap & Accessibility

- Many Bootstrap components include built-in accessibility features.
- Uses ARIA attributes to describe elements to screen readers.
- Provides proper keyboard navigation in modals, dropdowns, carousels, etc.
- Helps meet accessibility standards with less extra work.



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Alternative CSS frameworks

Bootstrap is widely used, but there are alternatives:

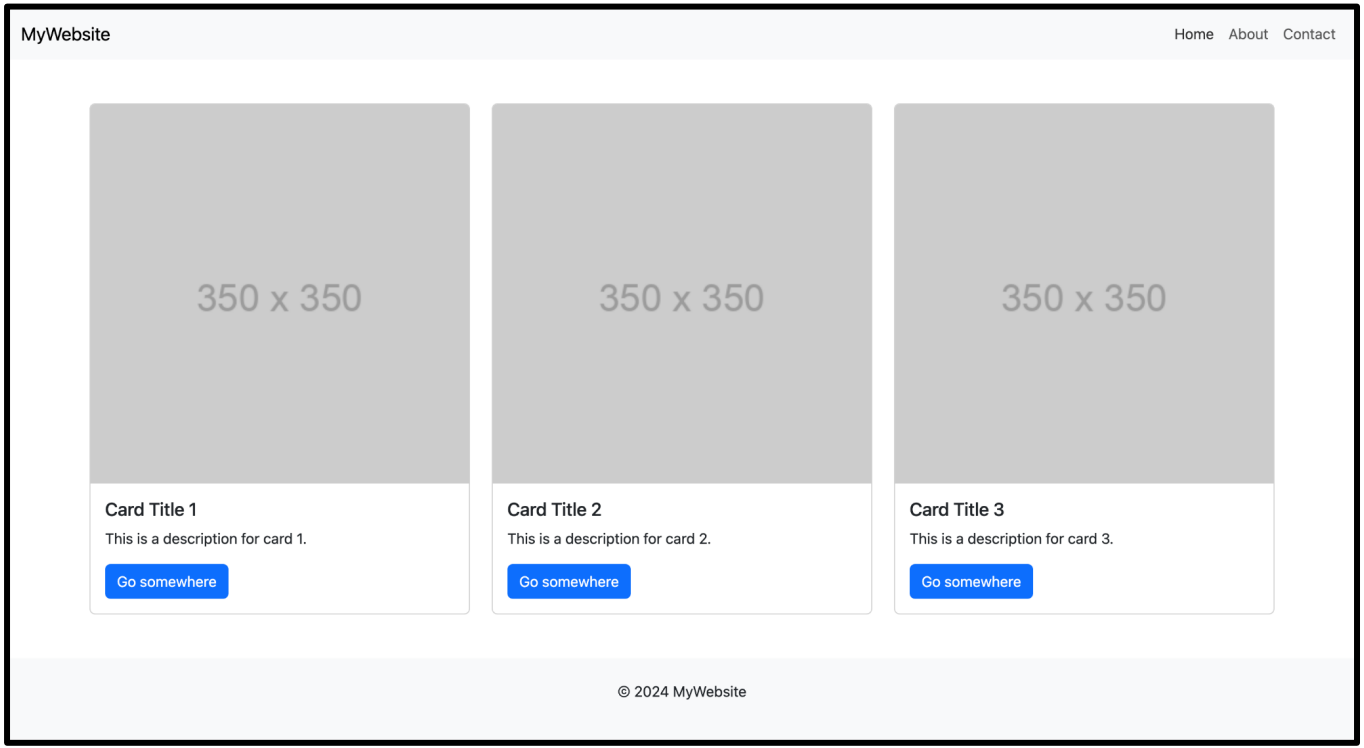
- Tailwind CSS (utility-first approach) tailwindcss.com
- Bulma (Flexbox-based) bulma.io
- Pure.css (lightweight) purecss.io
- Materialize (Material Design) materializecss.com
- Semantic UI (semantic class names) semantic-ui.com
- Foundation (enterprise-focused)

All are CSS frameworks — the choice depends on the project.
Worth knowing they exist for evaluating design options critically.

Summary

- Web accessibility makes sites usable for all, including disabilities, but benefits everyone (e.g., better contrast on mobile in sunlight)
- POUR principles: Perceivable, Operable, Understandable, Robust
- UI elements are reusable building blocks that build user trust through familiar patterns
- UI component library provides ready-made components for consistency and efficiency
- Bootstrap is
  - CSS framework with pre-built HTML, CSS and JS components
  - added via CDN or local files
  - used by adding CSS classes to HTML elements, the 12-column grid, and components
  - customizable with Bootswatch themes or your own CSS that overrides Bootstrap rules
  - widely used, but Tailwind, Bulma, Pure.css, Materialize, Semantic UI, and Foundation offer different approaches

Exercise: Implement the given layout



Bibliography

- Bootstrap documentation: https://getbootstrap.com/
- Bootswatch themes: https://bootswatch.com/
- WCAG Accessibility Guidelines: https://www.w3.org/WAI/WCAG21/
- MDN Accessibility: https://developer.mozilla.org/en-US/docs/Web/Accessibility
- Tailwind CSS: https://tailwindcss.com/
- Bulma: https://bulma.io/
- Pure.css: https://purecss.io/
- Materialize: https://materializecss.com/
- Semantic UI: https://semantic-ui.com/